



*Software Engineering (3) Course (SWE521): Agile
Software Engineering Skills*

AGILE SOFTWARE ENGINEERING SKILLS

LECTURE 02

Self-Organizing Teams

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Outline

1. Introduction
2. Self-Organizing Teams
3. Groups and Teams
4. Agile Principles
5. Forming Teams
6. Collaboration Activities Within Self-Organizing Teams
7. Virtual Teams



Introduction

- **Teams** : group of people come together to carry out a shared task.
- The norm for business information system development now is **Agile methods**
- **This chapter will answer the following questions:**
 - How teams are formed?
 - How we can enhance the performance of these teams?
 - What distinguishes groups from teams?



Self-Organizing Teams

- **Self-organizing teams function:**
 - manage their own achievement of tasks towards goal.
- The **customer** defines the **goal**.
- The **team** manages its **delivery**.
- The **priorities** around *what* must be accomplished come from the **business domain**,
- **Decisions** about *how* to achieve the goals come from within the **team**



Attributes of Self-Organizing Teams 1 to 7

1. **Autonomy**
2. **Requisite Variety**
3. **Learning to Learn**
4. **Cross-fertilization**
5. **Self-evaluation**



Attributes of Self-Organizing Teams 2 to 6

- **Autonomy**

- Self-organizing teams have the autonomy to manage and assume responsibility for their own **tasks** and **experience**
- **minimum** outside interference in their day-today activities
- **Only** the critical factors that are needed to direct the team are externally defined

- **Requisite Variety**

- The solutions used on a previous project might not work for this project.
- Self-organizing teams can be more adaptable at handling changing landscape.
 - Hence, self-organizing teams develop a **variety of approaches** to problem-solving, which they can deploy when needed.



Attributes of Self-Organizing Teams 3 to 6

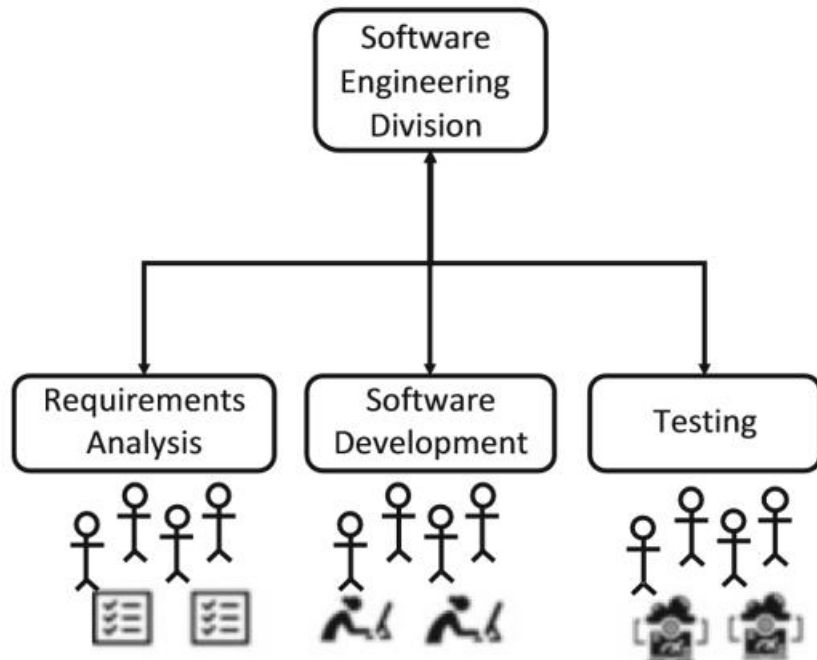
- **Cross-fertilization**

- **Conventional** organization form teams into organizational units comprising people with the **same skill** set
- Self-organizing teams, in contrast, are composed of individual members with diverse skills and specializations,
- Agile software development teams include **developers**, **testers** and business **analysts**. Multidisciplinary (multi-specializations) teams foster *cross-fertilization* within and also between teams.

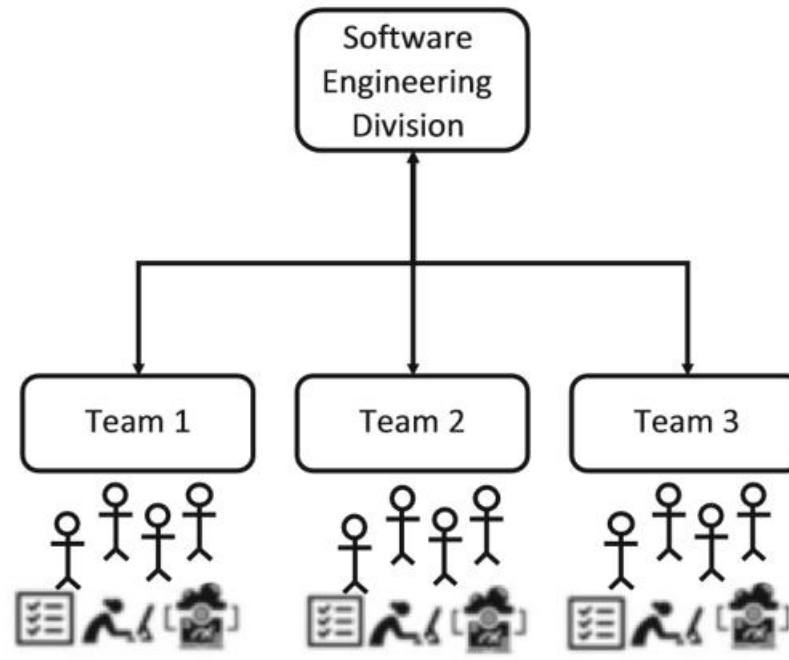


Attributes of Self-Organizing Teams 4 to 6

(a) Functional Teams



(b) Multi-disciplinary Teams





Attributes of Self-Organizing Teams 5 to 6

- **Learning to Learn**

- Teams learn to learn when they change the way they look at problems, review and refine the best work **method** and reconsider the best outputs to deliver.
- Agile practices that help teams learn to learn include **retrospective workshops** and **standup meetings**, more on these practices



Attributes of Self-Organizing Teams 6 to 6

Self-evaluation

- Self-organizing teams **establish** their own **goals** and **keep** on ***evaluating*** themselves such that they are able to **innovate** newer and better **ways** of **achieving those goals**.
- This **self-evaluation** is often **aimed** at improving **productivity** and product **quality**.
 - We find that productivity and product quality are often **conflicting** objectives, and **achieving the right balance** between them requires constant attention.



Groups and Teams 1 to 2

1. Team size (7 member +2, -2)

Larger groups of people **have difficulty** interacting constructively as a team

2. Skills portfolio

- Teams are used to bring together group members with **complimentary skills**
 - Teams where everyone **has exactly** the same skills are **less effective** than teams with a **mix of skills**

3. Common purpose (including shared performance or quality goals)

- Teams are **created** around a **common purpose** or a **shared set of goals**.
- A meaningful purpose **establishes aspiration** for the team.
- Often teams are **characterized** by **shared commitment** to a set of **performance** or **quality goals**



Groups and Teams 2 to 2

4. Common approach

- teams are committed to a **common approach**.
- This is the way they will work together to achieve their shared goals

5. Mutual accountability

- team members are jointly accountable for **outcomes**.
- A team also shares responsibility for its own successes and failures.



Building Team Performance 1 to 2

- Some techniques that can help build team performance
 - Establish urgency and direction.
 - Select members based on skills and potential (**not personalities**).
 - Pay attention to first meetings and actions.
 - Set rules for acceptable behavior.
 - Focus on few short-term performance tasks and goals.
 - Update facts regularly.
 - Spend (**lots of**) time together.
 - Exploit positive feedback.



Building Team Performance 2 to 2

- **Team members need to specialize in each of these three areas:**
 - Technical **skills**
 - Problem-solving **skills**
 - Interpersonal **skills**
- **Being more focused on delivery of working code, you might want to make sure you have:**
 - **Front-end**, human-computer interaction and user experience skills
 - **Back-end**, web services and database implementation skills
 - **Testing** and evaluation skills
 - **Deployment skills** to place working code on servers accessible by users



Start-up Team Members 1 to 2

Software Company Start-up Team Members

In software company start-ups, there is a trendy saying that your team should comprise a *hacker*, a *hipster* and a *hustler*. The *hacker* is an innovation thought leader and brings strong technical skills. The *hipster* brings graphical design, visual creative and user experience skills, while the *hustler* is a dealmaker and negotiator who can close sales. Okay, so this saying is a bit of cliché, but it illustrates the benefits of a diverse skill set in your teams.



Start-up Team Members 2 to 2

- **The set-up phase of the group is very important.**
 - Purposeful and collaborative atmosphere is to be encouraged.
 - Setting the right tone at early stages is healthy and important.
 - The team needs to gain consensus on working practices and acceptable behavior.
 - You need to **identify** tasks you can complete and improvements you **can make** with **modest effort**, short time-scales and measurable success.
 - We need to **ensure** that the team benefits from regular updates of information.
 - In self-organizing agile team, we use daily coordination meetings, in scrum called stand-ups, to update everyone on status.
 - Face-to-face time together can help build trust and mutual confidence.
 - Feedback to the team needs to reinforce the growth, goals or performance objectives



Agile Principles 1 to 3

- Iterative and incremental development
- We are interested in the principles of
 - sustainable pace
 - and collective code ownership.



Agile Principles 2 to 3

• Sustainable Pace

- It is not good idea when your boss, through omission or design, prevails on you to work all night.
- **This realization**, that all-night coding is not ideal, partly informs the concept of sustainable pace.
- The idea is that software engineering is **a creative activity** that should be **conducted** when people **are awake**, alert and fully focused on the job.
- This is the idea that creative work needs to be conducted in normal office working hours and not involve long periods of evening or weekend working.
- The implication is that carefully implemented software development processes enable a **sustainable approach** to code creation, **sustainable** over weeks, months and years.



Agile Principles 3 to 3

• Collective Code Ownership

- Code can be written by individuals but, according to the concept of collective code ownership, should be a resource belonging to the whole team.
- It is argued that we should not put names on the modules of software we write.
- There should be no obstacle to making changes or corrections to code written by others.
- The team, as a whole, stands or falls by the software created by its members.



Forming Teams 1 TO 6

- The most well-known process that happen when people come together is comprising:
- **Forming:**
 - Group members create a team with clear structure, goals, direction and roles so that members begin to build trust
- **Storming**
 - During the storming phase, frustrations and perhaps conflicts build up. The team often needs to refocus on its goals, perhaps breaking larger goals down into smaller, more achievable steps
- **Norming**
 - team members begin to resolve the differences between their expectations and the reality of the team's experience.



Forming Teams 2 TO 6

- **Performing**

- During the **performing phase**, there is **significant progress** towards team **goals**, and team members **show high commitment** to the team's **goals**.

- **Adjourning (postpone) stages**

- during the **adjourning phase**, team members **complete** their **deliverables** **evaluate performance** with a particular **focus** on identifying '**lessons learned**' and **celebrate** the **contributions** and **accomplishments** of the team.
 - **Deliverables**: final software, test execution, reports



Forming Teams 3 TO 6

- **Accelerating Team Formation:**

- During **the early stage** of coming together as a group, we can **undertake** several **activities** to help establish **shared goals** and **build trust**.



Forming Teams 4 TO 6

- ***Handling Difference and Conflict***

- There are **three main attributes** of techniques for handling disagreements. When negotiating solutions to differences of opinion:
 1. *Focus on the problem.*
 2. *Avoid focusing on personalities.*
 3. *Seek solutions that maximize benefit to more of the participants*
- Taking a **vote** could seem like a good solution, but the majority within the team may not be as well informed as an expert.
 - So **discussion and learning**, while working towards consensus, is often a better way to reach agreement.



Forming Teams 5 TO 6

- **Accelerating Norming**

- Successful team norming **relies** on
 - **good communication**
 - and a **shared sense** of purpose within the group.
- Team members are focused on the task at hand, and not on
 - conflict resolution
 - and impediments (obstacles) to efficient working



Forming Teams 6 TO 6

Theory X and Theory Y

It has been observed that there are two views of organisations . Theory X describes organisations founded on the idea that people don't want to work and need to be coerced or controlled to create good work. Theory X advocates a hierarchical approach to management structures, where roles are task oriented, repetitive and inflexible.

Theory Y, in contrast, suggests that people want to work to gain self-esteem, recognition and satisfaction. In Theory Y, the role of management is to create a supportive environment in which people can do their best work. There is less emphasis on command and control structures and more focus on facilitating opportunities for collaboration.



Collaboration Activities Within Self-Organizing Teams 1 TO 7

- These activities may be performed by **different team members** or by the same team member at different times.
- An **attribute** of more **experienced** and adept **team members** could be the ability to perform more of these activities on behalf of the team.

Role	Function
Mentor	Guides teammates (members), encourages Agile practices
Coordinator	Manages customer interactions
Translator	Clarifies business language for technical understanding
Champion	Advocates Agile to leadership
Promoter	Engages customer collaboration
Terminator	Addresses long-term disruptions within the team



Collaboration Activities Within Self-Organizing Teams 2 TO 7

- **Mentor**
- The **scrum master** helps **inculcate** (**enhance**) the use of agile methods in the team.
 - But other members of self-organizing teams must also **mentor each other**.
 - Perhaps **someone** new **joins** the team and needs **advice** to get started and **feel** welcome.
 - Perhaps **someone** already in the team **needs** to learn a new technology.
 - Mentors **guide** and **support team members**, **help** them become more confident about using agile methods and **encourage** the ongoing use of agile practices.



Collaboration Activities Within Self-Organizing Teams 3 TO 7

- ***Coordinator***

- The coordinator **acts** as a **representative** of the team to **manage** customer **expectations** and **coordinate customer collaboration** with the team.
- The coordinator has to **collect** team **member requests** for information from **customers** (to avoid irritating duplication of questions) and deal with the whole issue of **change request handling**



Collaboration Activities Within Self-Organizing Teams 4 TO 7

- **Translator**

- **Translators** are needed that can make clear the meaning of business language used by customers for the benefit of technical team members.
- **Where Miscommunication Happen?**
 - **Business Focus:** Outcomes like customer retention, user satisfaction, and marketing impact.
 - **Technical Focus:** Implementation details like data models, API endpoints, and performance constraints.
- **Participants** in the self-organizing team need to contribute to improved communication between these two domains.
- Understanding the business domain of the project is the **primary role** of the **product owner**.
 - But self-organizing teams benefit from gaining this understanding too.



Collaboration Activities Within Self-Organizing Teams 5 TO 7

• Champion

- **Champions:** are team members that **advocate** for agile methods with senior management within their organization.
 - The champion is adept at explaining agile benefits using language and evidence that is convincing for **senior executives**.
- We want **senior executive** support for the self-organizing agile team.
- It is **hard for agile** to flourish **without** senior management support.



Collaboration Activities Within Self-Organizing Teams 6 TO 7

• Promoter

- The **promoter** is a **proponent** (support) of agile methods with customers.
- The promoter **secures** customer involvement and collaboration to **support** the efficient functioning of the self-organizing team.
- Customers **play** a **vital role** in **identifying** and **prioritizing** requirements, while the promoter **ensures** that the team **gets** all the support needed from customers.



Collaboration Activities Within Self-Organizing Teams 7 TO 7

• Terminator

- Self-organizing teams are often happy to take damage limitation steps to cover for a team member who is 'having a bad day'.
- But if a team member is causing problems over a long period, then more drastic action may be needed.
- In the most extreme case, members of the self-organizing team may engage external stakeholders to get support for removing someone from the team.



Virtual Teams 1 to 7

- Virtual teams are **formed**, when some or all team members work **remotely** and **communicate** using technology.
- The time, cost and environmental impact of daily travel to work is avoided.



Virtual Teams 2 to 7

- **Principles for Virtual Team Management**

- It is particularly important to **ensure** there is **agreement** on **clear goals** with virtual teams.
- Communication and collaboration **processes** are needed to expose and resolve **misunderstandings** and diffuse conflict.
- Communicating as a group on different topics and activities and building awareness of other team members **help** with **goal achievement**.
- Helping team members **build inter-dependence** through **goal setting** and **task design** **help** with the development of successful teams.
- Trying to **help** team members **understand** the **strengths** each person brings to the work **helps** overcome disconnectedness.



Virtual Team Collaboration 3 to 7

- Benefits: Remote work, time savings, broader talent pool
- Challenges: Miscommunication, isolation, lack of spontaneous (**automatic**) bonding
-  *Virtual Success Principles*
 - Align goals and responsibilities
 - Use collaboration tools wisely (Slack, Zoom)
 - Foster team spirit through shared social time



Virtual Teams 4 to 7

- **Preparation for Team Success**

1. **Cultural Diversity**

- Cultural diversity is a feature of globally distributed teams
- Cultural ambassadors with experience and understanding of the different cultures can support **communication** at times of **stress** or **tension** within the group.
- Cultural **awareness** and **sensitivity** are needed for group members to empathize and build **trust**



Virtual Teams 5 to 7

2. Remote Pair Programming

- Tactical use of remote pair programming can be useful tool for managing **difficult tasks**.
- Set aside time for **pair programming** during **sprint planning**.
- Each pair programming session **needs clear goals** and a **set agenda** before the start.
- Pair programming also is a good way to **on-board new members** of a team.
- A basic way of doing remote pair programming is to use any video conferencing platform that allows screen sharing.
- Customary approaches include:
 - **Driver/navigator**: the **driver** works on careful implementation of code, while the **navigator** is doing a real-time code **review** while thinking about issues such as **readability** and **architecture adherence**; **swap every 30 min or so**.
 - **Ping-pong**: One person writes a test, while the other person then codes against the test criteria. Once the test passes, **swap roles**



Virtual Teams 6 to 7

- **Launch**

- The launch phase of a virtual team is particularly important.
- A *kick-off* phase serves five main purposes:
 - Getting to know other **team members**
 - Clarifying the **team goals**
 - Clarifying team member **roles** and **functions**
 - Discussing efficient communication **technologies use**
 - Developing general **teamwork rules**



Virtual Teams 7 to 7

- ***Performance Management***
- Virtual teams are **formed**, when some or all team members work **remotely** and **communicate** using technology.
- The time, cost and environmental impact of daily travel to work is avoided.



Communities of Practice

- It is useful to be aware of **another type** of team that **exists** within the software development ecosystem.
- These are **teams** with members that **do not contribute** to **shared work tasks** but that nevertheless **share ways of working**, **membership rituals** and often **shared goals**
- A community of practice is a **voluntary**, often rather **unstructured**, group that **supports** and **facilitates knowledge** and **experience sharing**.
- In the agile development culture, the concept of **Guilds** is introduced that to formalize the community of practice as part of the development process.
 - The Guilds tend to be organic and emergent.
 - Guilds **vary** in **size**, **mission**, **membership** and **activities**.



Summery

- Most people engaged in a software project will be members of self-organizing teams.
- In this chapter, I have explained some defining characteristics of teams and some techniques to encourage higher performance from teams.
- I've focused on team life cycle and team formation issues.
- I've focused on some research, which shows what collaboration activities self-organizing teams in software projects undertake.



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END OF LECTURE 02

Self-Organizing Teams

NEXT



Lecture 02: Self-Organizing Teams (Exercises)